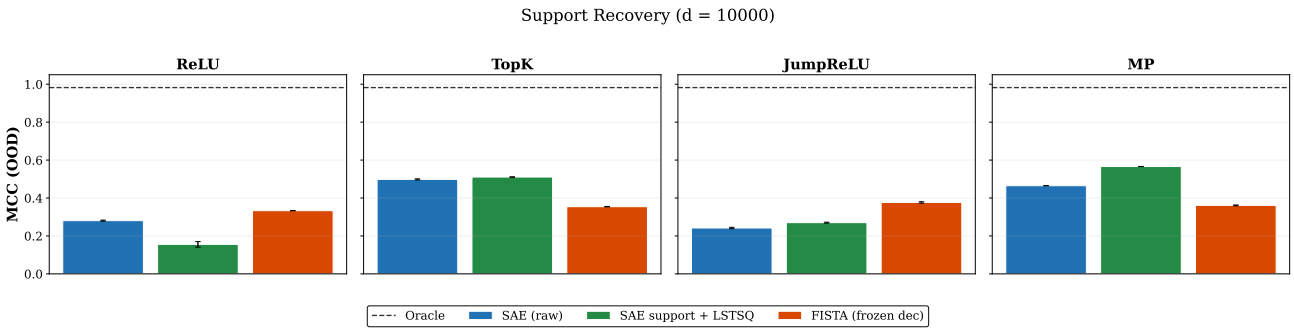

Figure 57: Support recovery, $d_z = 10000$.

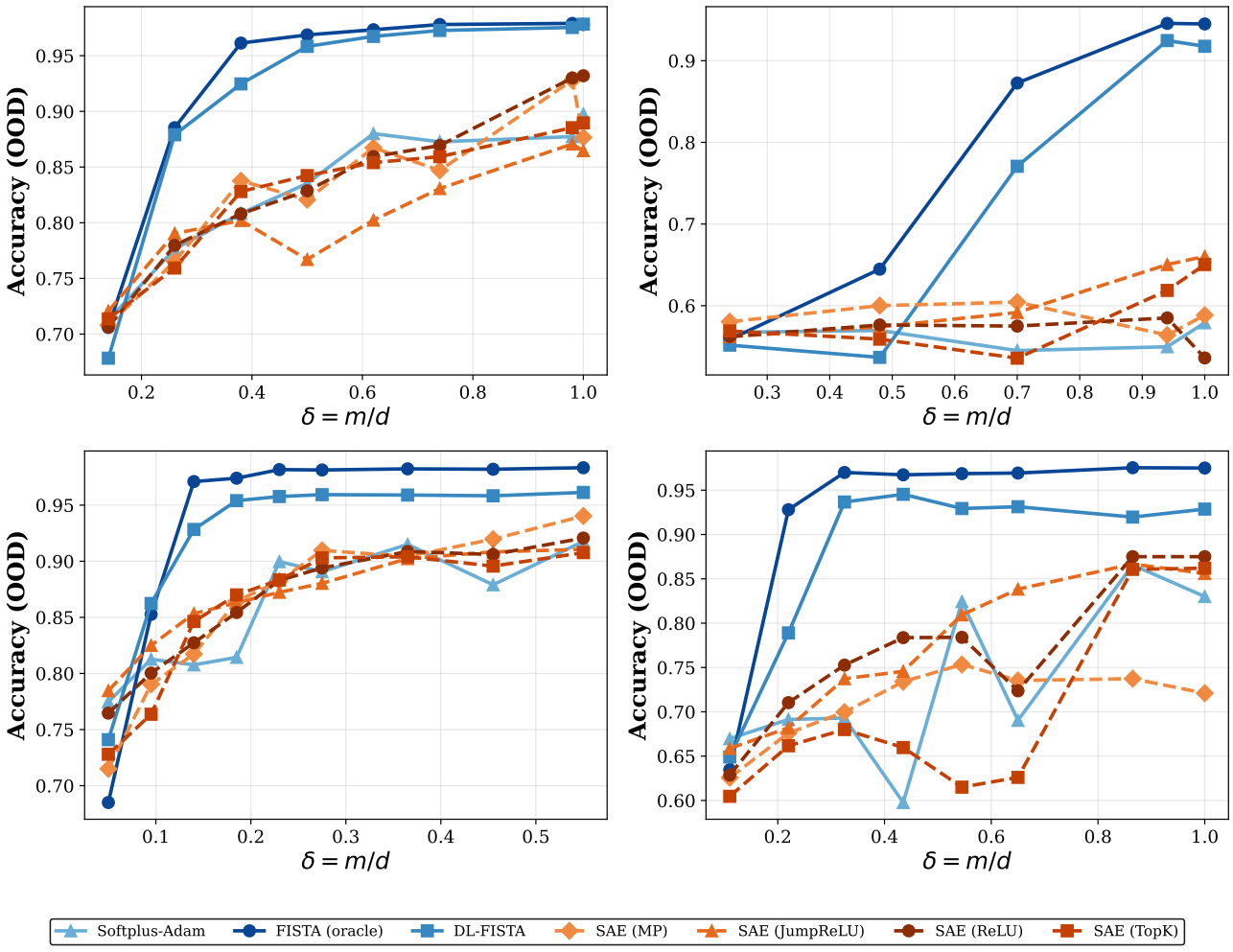

Figure 58: **Phase transition: Accuracy (OOD).** SAE OOD accuracy plateaus while per-sample methods transition sharply. The gap is most severe at moderate δ where compressed sensing succeeds but amortised inference fails.

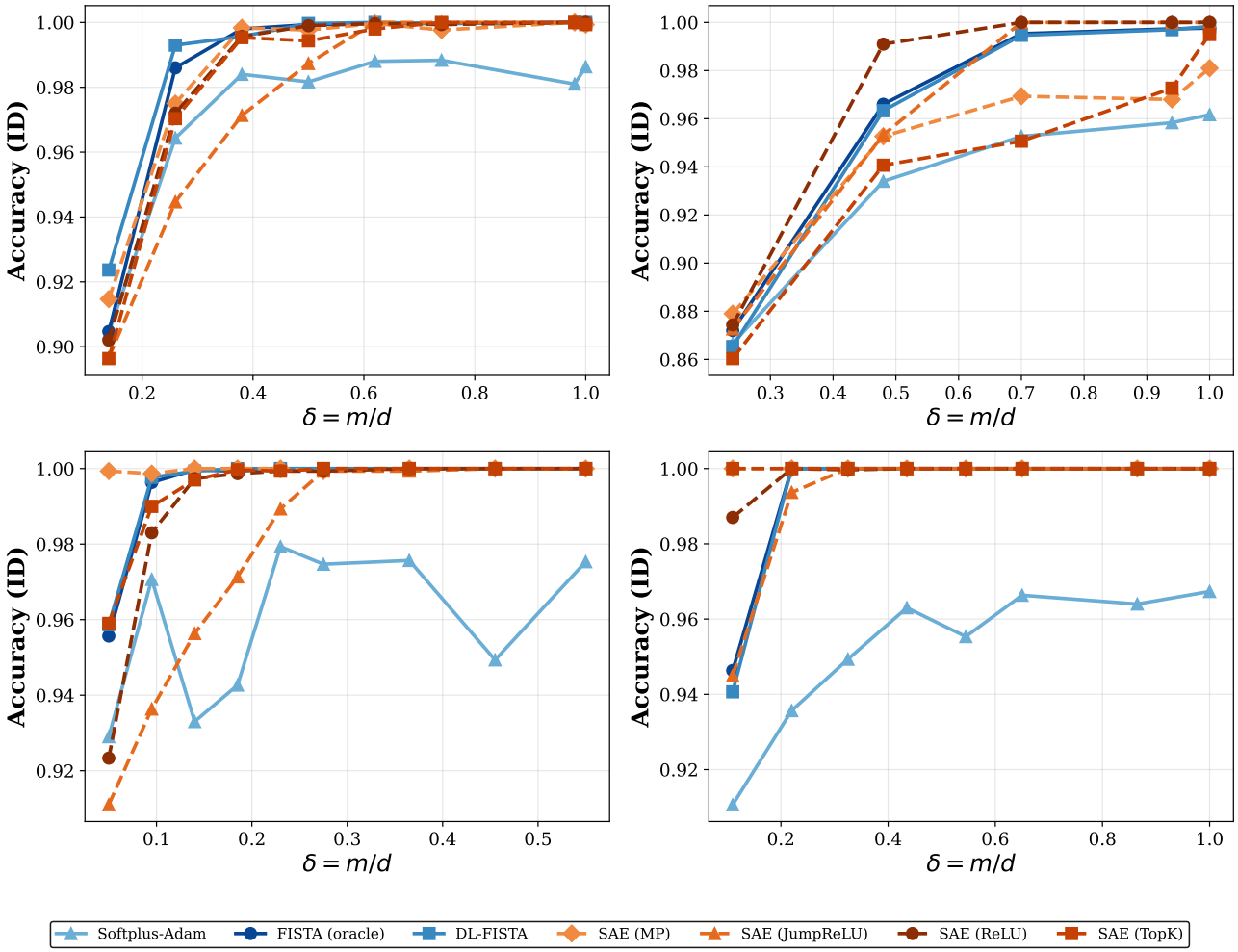

Figure 59: **Phase transition: Accuracy (ID).** ID accuracy is high for most methods once δ is sufficient, but SAEs show more variance and lower peak accuracy than per-sample methods.

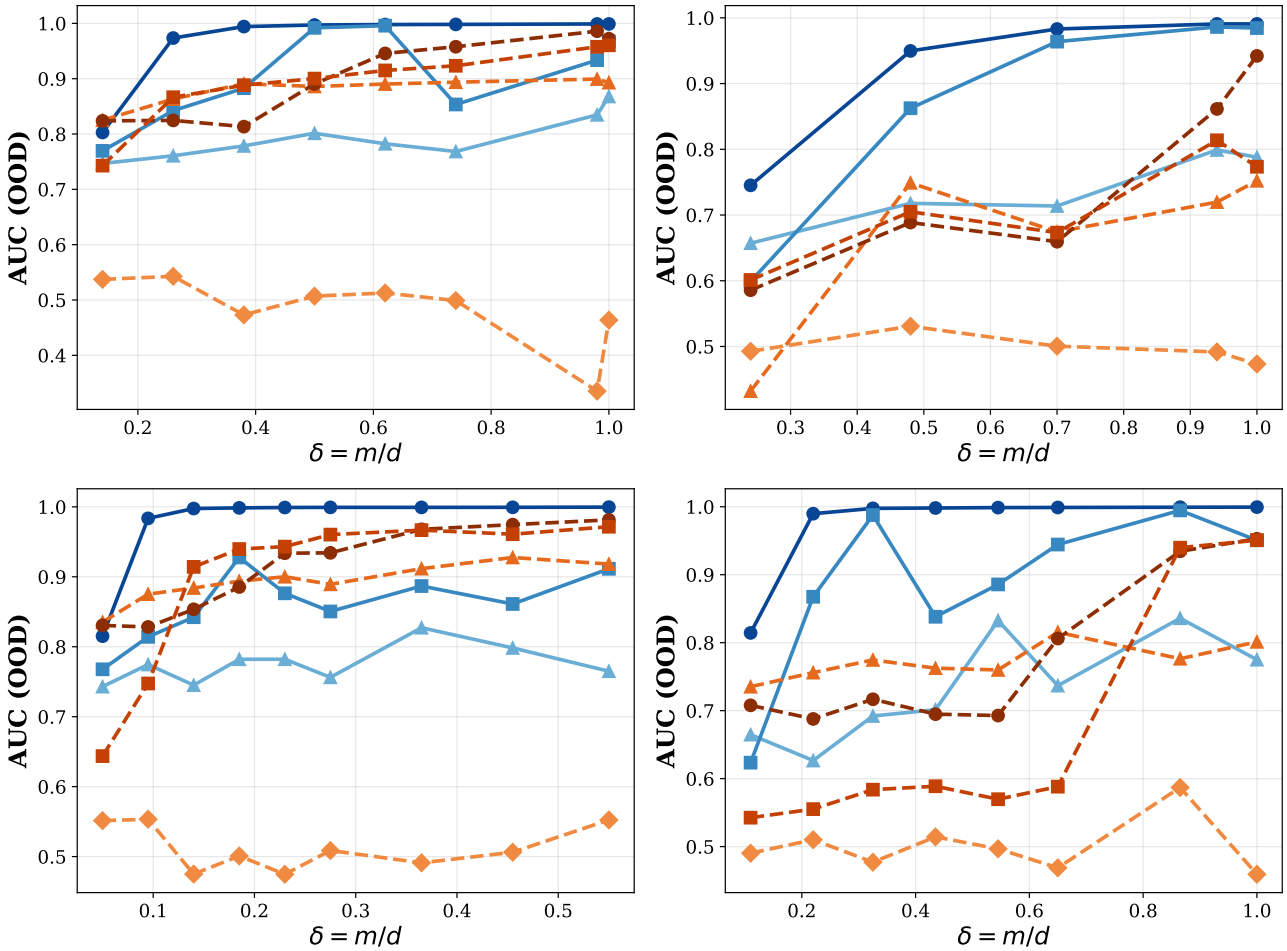

Figure 60: **Phase transition: AUC (OOD).** Per-feature AUC on OOD data confirms the same pattern: per-sample methods achieve near-perfect AUC while SAE features fail to isolate the label under novel compositions.